

Understanding Profit and Delivery Challenges in Supply Chains with PowerBI

When I first started this project, I was curious about the factors affecting supply chain efficiency. I remember watching a documentary about how shipping delays can impact businesses, and it stuck with me. Who knew that delving into data from Sword Consulting Group would lead me to some surprising insights about customer spending and delivery challenges?

Why THIS Project?

My motivation for choosing this project came from a blend of personal interest and the relevance of supply chain management in today's business landscape. Seeing how companies struggle with late deliveries and customer satisfaction pushed me to explore this area. I wanted to find tangible solutions using data analysis, and PowerBI offered the perfect tools for the job.

What Readers Will Gain

In this article, I aim to share my findings regarding customer spending patterns and delivery factors. You'll learn about:

- The most profitable products.
- The surprising lack of influence that price and product category have on shipping times.

Key Takeaways

- Customers spend the most on computers and electronics.
- Fishing products, while not as frequently purchased, are more expensive overall.
- Price and product categories do not affect shipping speed.

Dataset Details

The data I worked with comes directly from Sword Consulting Group, and it is extensive, containing 180,519 entries. This large dataset was ideal as it provided a comprehensive view of customer buying behaviors across various product categories. Having such a robust dataset helped ensure the results were more reliable and gave me a wider perspective on trends.

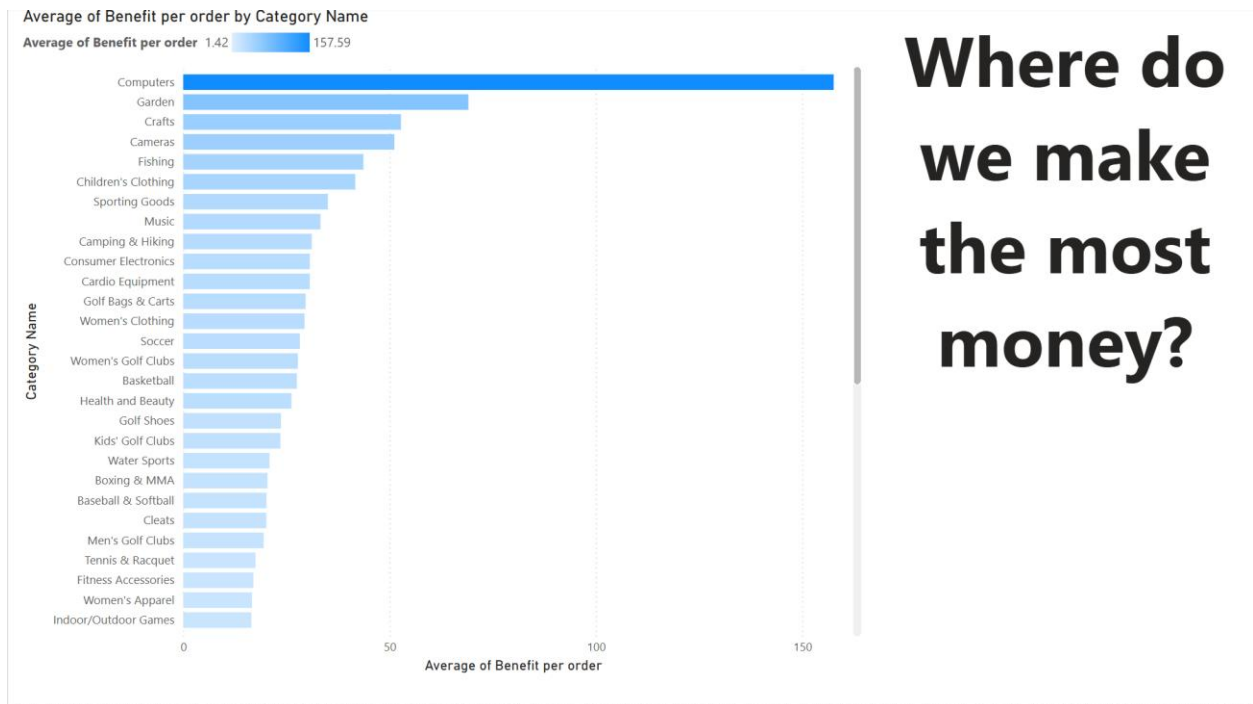
Analysis Process

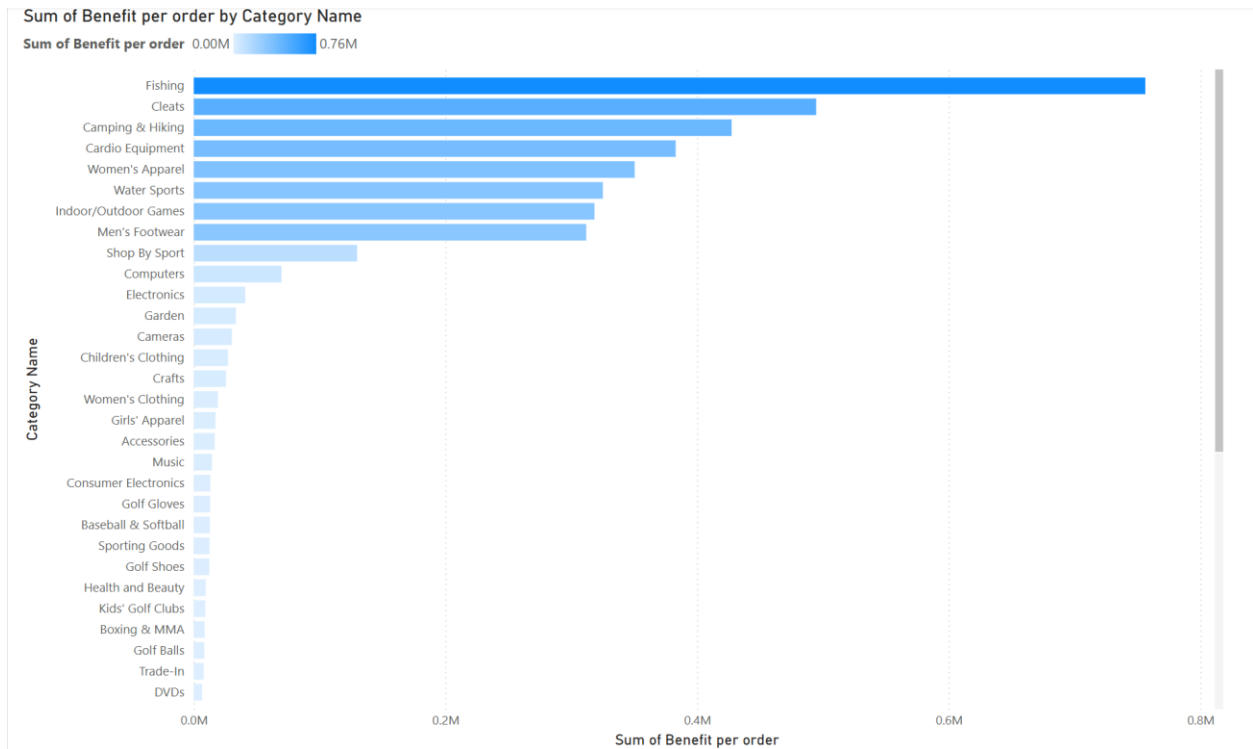
In my analysis, I followed several steps:

1. **Data Cleaning:** I ensured the data was accurate and free from errors.
2. **Transformation:** I structured the data to focus on key metrics like average spending and shipping times.
3. **Visualization:** Using PowerBI, I created visuals that highlighted spending patterns and delivery times.
4. **Statistical Methods:** I examined relationships between prices, product categories, and shipping speed.

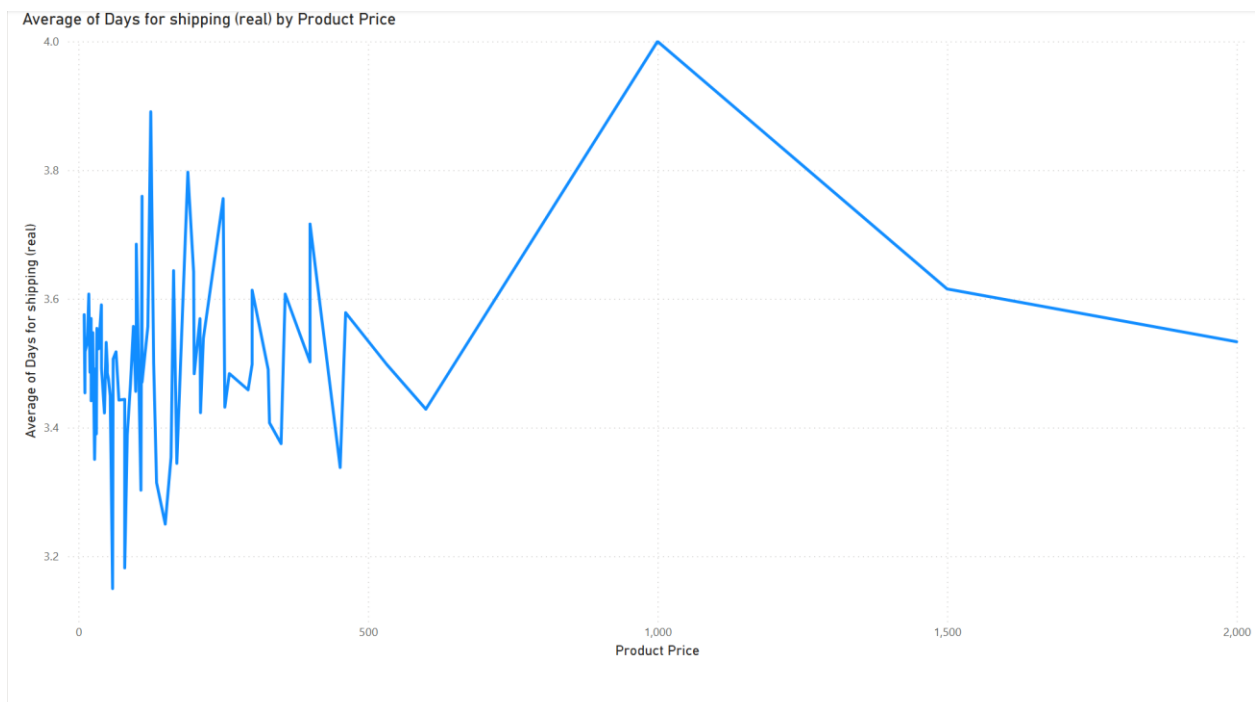
I was surprised to find that neither price nor product category influenced shipping times. It led me to wonder if other factors, like logistics or carrier performance, were at play.

Visuals and Insights

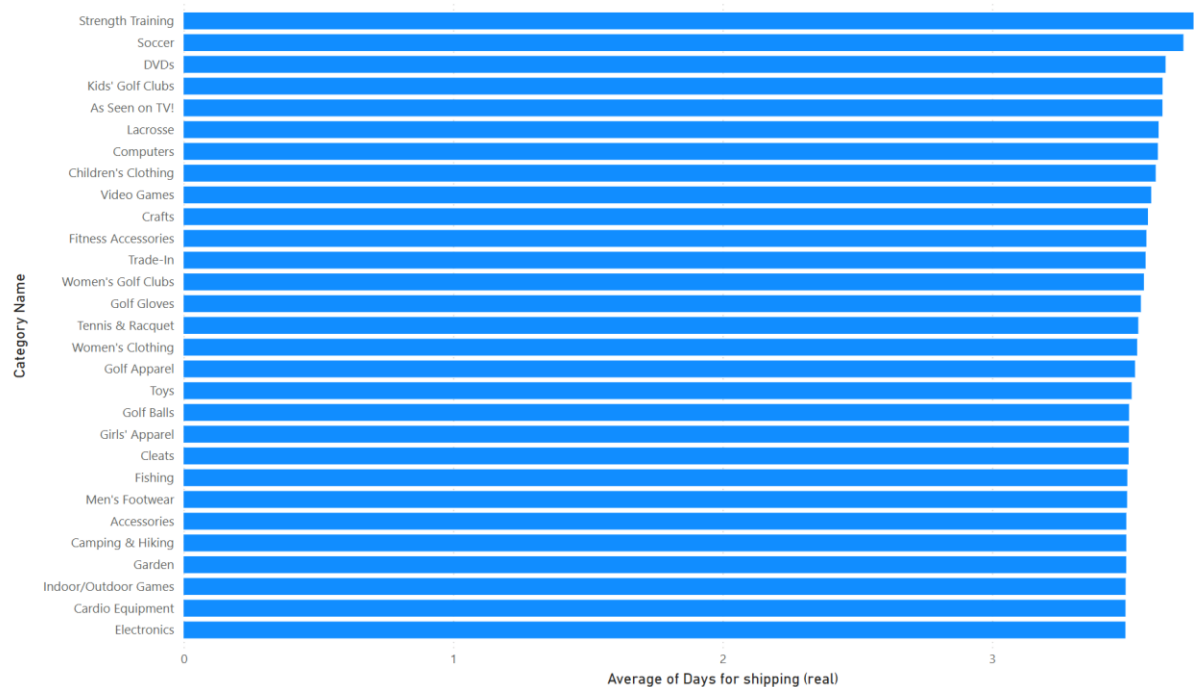




One of my key visuals shows the average amount spent per product category. Here, you can see that customers spend the most on computers and electronics, which is no surprise given the tech-infused world we live in. However, the total for fishing products is surprisingly high, indicating that while fewer items may be sold, those that are purchased carry a hefty price tag.



Average of Days for shipping (real) by Category Name



The other graphs illustrate the impact of price and product category on shipping speed. Interestingly, there was no if maybe a slight correlation found, suggesting that shipping delays might be attributed to other logistical issues instead.

Main Takeaways

- The highest average spending was on computers and electronics, indicating a focus for profit strategy.
- Fishing products, while not as commonly bought, generate higher revenue per sale, shedding light on niche markets.
- The lack of relationship between price, category, and shipping suggests that businesses need to look beyond these factors to resolve delivery issues.

Conclusion and Personal Reflections

This project taught me a lot about the complexities of supply chains. One significant challenge I faced was interpreting the data accurately, as the vast amount of information could be overwhelming. However, breaking it down into manageable parts helped me focus on the most relevant insights.

I now have a greater appreciation for the nuances of customer behavior and the factors affecting delivery performance. Moving forward, I plan to explore how logistics and operational strategies can further optimize supply chain efficiency.

Call To Action

I invite you to connect with me on LinkedIn! If you or someone you know is looking to hire a data analyst, let's talk! I'd love to hear your thoughts or questions about this project.